

LABORATORY RECORD

Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	2
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Date	30-08-2026

1. TITLE

Design and Implementation of a Styled Webpage Using HTML5 and CSS3

2. AIM

To design and implement an attractive static webpage using HTML5 and CSS3 with navigation links, content cards, gradients, text shadows, hover effects, flexbox layout, borders, rounded corners, buttons, and a footer.

3. OBJECTIVE

1. To create a structured webpage using HTML5 elements.
2. To link an external CSS file with an HTML document.
3. To apply CSS properties such as background, color, padding, margin, and font styles.
4. To arrange webpage cards using CSS Flexbox.
5. To implement hover effects for navigation links and buttons.
6. To create visually attractive cards using borders, border-radius, and box-shadow.

4. THEORY

CSS3 (Cascading Style Sheets) is used to control the presentation and layout of HTML elements. CSS can modify colors, fonts, spacing, borders, shadows, and alignment. Flexbox provides an efficient way to arrange elements horizontally or vertically. Pseudo-class selectors such as `:hover` allow styles to change when the user interacts with an element. External stylesheets keep webpage structure and presentation separate.

Application Flow: HTML5 Structure → External CSS → Navigation Styling → Flexbox Cards → Hover Effects → Footer

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and define the page title as 'Technology Hub'.
2. Link the external stylesheet `style.css` inside the head section.
3. Create a header containing the main webpage heading.
4. Create a navigation bar with Dashboard, Projects, Courses, and Support links.
5. Create three content boxes for Python, JavaScript, and Database topics.
6. Add headings, descriptions, and Explore buttons inside each box.
7. Create a footer displaying the copyright information.
8. In CSS, style the body, header, navigation, links, content area, boxes, buttons, and footer.
9. Use Flexbox to align the content boxes and add hover effects to links and buttons.
10. Open the HTML file in a browser and verify the layout and styling.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Technology Hub</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>Technology Hub</h1>
</header>

<nav>
<a href="#">Dashboard</a>
<a href="#">Projects</a>
<a href="#">Courses</a>
<a href="#">Support</a>
</nav>

<section class="content">

<div class="box">
<h2>Python</h2>
<p>Python is a popular programming language used for software development.</p>
<button>Explore</button>
</div>

<div class="box">
<h2>JavaScript</h2>
<p>JavaScript adds functionality and interactive features to websites.</p>
<button>Explore</button>
</div>

<div class="box">
<h2>Database</h2>
<p>Databases are used to store, organize, and manage application data.</p>
<button>Explore</button>
</div>

</section>

<footer>
<p>© 2026 Technology Hub</p>
</footer>

</body>
</html>
```

style.css

```
body {
margin: 0;
font-family: Arial, sans-serif;
background: #eef5f9;
}

header {
background: linear-gradient(to right, #ff6600, #ff0066);
color: white;
text-align: center;
padding: 25px;
text-shadow: 2px 2px 5px gray;
}

nav {
background: #222;
text-align: center;
padding: 15px;
}

nav a {
color: white;
```

```
text-decoration: none;
margin: 20px;
font-size: 18px;
}

nav a:hover {
color: orange;
}

.content {
display: flex;
justify-content: center;
gap: 25px;
margin: 35px;
}

.box {
background: white;
width: 250px;
padding: 20px;
border: 2px solid #555;
border-radius: 10px;
text-align: center;
box-shadow: 5px 5px 10px #999;
}

button {
background: #0066cc;
color: white;
border: none;
padding: 10px 18px;
border-radius: 5px;
cursor: pointer;
}

button:hover {
background: #ff6600;
}

footer {
background: #222;
color: white;
text-align: center;
padding: 15px;
margin-top: 20px;
}
```

7. CODE EXPLANATION

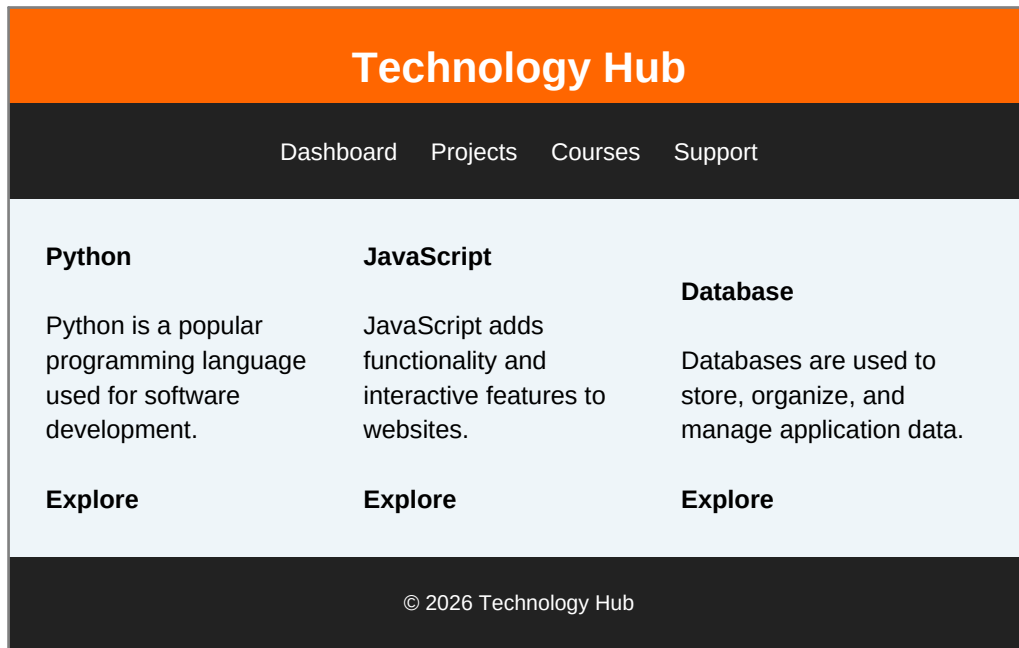
Code / Syntax	Explanation
<header>	Defines the header section containing the main webpage heading.
<nav>	Creates the navigation section containing webpage links.
.content { display: flex; }	Uses Flexbox to arrange the content boxes in a row.
.box	Styles each content card with padding, border, rounded corners, and shadow.
:hover	Changes the appearance of links and buttons when the mouse moves over them.
linear-gradient()	Creates a smooth color transition in the header background.
box-shadow	Adds a shadow effect around the content boxes.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open webpage	Header and navigation are displayed	Displayed correctly	Pass
TC-02	Inspect cards	Three topic cards appear in a row	Cards aligned using Flexbox	Pass
TC-03	Move mouse over link	Link color changes	Hover effect works	Pass
TC-04	Move mouse over button	Button background changes	Hover effect works	Pass
TC-05	Inspect card styling	Border, radius and shadow appear	Styling displayed correctly	Pass

9. OUTPUT

Output 1: Styled Technology Hub webpage with navigation, cards, hover effects, buttons, and footer.



10. RESULT

Thus, the styled Technology Hub webpage was successfully designed and implemented using HTML5 and CSS3. The webpage was verified for navigation styling, Flexbox layout, card design, hover effects, buttons, and footer presentation.